

# Bhavika Jata

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AI & Data Science engineer (B.Tech in Artificial Intelligence and Data Science, latest SGPA: 10/10) specialising in production-ready GenAI systems, RAG pipelines, LLM-powered agents, and FAISS-backed vector search. Engineered an NLP-to-SQL AI agent at Tata Technologies and automated data pipelines at L&T Precision cutting manual effort by 40%. Built and deployed ResearchEngine, a multi-module AI research assistant with semantic document ingestion and mentor-style LLM conversations. Targeting AI/ML engineering and GenAI product roles.

## WORK EXPERIENCE

### AI Intern

Tata Technologies Ltd., Pune

Dec 2025 – Jan 2026

- Built an NLP-to-SQL AI agent for the HR department using LangChain (via Langflow), enabling non-technical staff to query databases in plain English.
- Architected 3+ modular LangChain agent workflows, reducing prototype-to-demo time by ~30% and cutting experimentation cycle time by 25%.
- Engineered reusable Python-based AI components, improving workflow modularity and enabling faster iteration.

### Data Analyst Intern

L&T Precision Engineering and Systems, Mumbai

May 2025 – Jun 2025

- Automated Python-based web scraping pipelines, cutting manual data collection effort by 40%.
- Built an SVM-based sentiment analysis model attaining 78.0% classification accuracy.
- Crafted an interactive Power BI dashboard adopted by business teams to track KPIs and surface weekly actionable insights.

## EDUCATION

### K. J. Somaiya School of Engineering, Mumbai

B.Tech in Artificial Intelligence & Data Science (Honors in Data Engineering)

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Database Management Systems, Probability & Statistics

Aug 2023 – May 2027

SGPA: 10/10 | CGPA: 8.908/10

## PROJECTS

### ResearchEngine

May 2026 (ongoing)

- Developed an AI-powered RAG-based LLM research assistant using Python, LangChain, Gemini, and FAISS, deployed using Netlify for live use beyond local development.
- Built a multi-format document ingestion and semantic search system for research documents.
- Designed for researchers at all levels — from literature discovery and paper analysis to mentor-style AI conversations, significantly reducing multi-source search effort.

### CNN vs. Quantum CNN — Sugarcane Leaf Disease Detection

Jan 2026 (ongoing)

- Benchmarked Custom CNN, EfficientNet-B0, and MobileNet-V2 on a 13-class sugarcane disease dataset, with EfficientNet-B0 achieving the highest test accuracy of 96.79% and F1-score of 0.97.
- Built Python preprocessing and augmentation pipelines training on 12,000+ images, with Custom CNN from scratch achieving 94.9% train and 93.21% validation accuracy as a strong classical baseline.
- Extending the study toward Quantum CNN architectures to evaluate whether quantum feature extraction can surpass EfficientNet-B0's 96.79% benchmark targeting a classical vs. quantum comparative analysis.

### Crop Recommendation & Yield Prediction System

Nov 2023

- Engineered ensemble ML models (Random Forest, XGBoost, Voting Classifier) in Python achieving 99.52% prediction accuracy on a scalable pipeline processing 10,000+ records across 21 crop types.
- Integrated MODIS satellite imagery via Google Earth Engine for geospatial temperature analysis.

### Resume Eligibility Parser

Jun 2024

Designed a rule-based NLP engine in Python processing 50+ resumes, eliminating manual review for 80% of unqualified applicants; deployed via Streamlit, cutting decision time from 5 minutes to under 30 seconds.

### Social Media Sentiment Analysis

Apr 2023

Built an end-to-end NLP pipeline using SVM (Python/Scikit-learn) achieving 92% accuracy classifying 2,000+ Google Form survey responses on social media usage, cutting collection and cleaning time by 60%.

## SKILLS

- Programming:** Python, SQLite, PostgreSQL, MS Excel
- Computer Vision:** Image Classification, Image Processing (OpenCV, Albumentations), CNNs, RNNs, LNNs Transfer Learning (EfficientNet, MobileNet-V2, ResNet), Data Augmentation
- ML / DL Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, SVM, Random Forest, CNN, RNN, Transformer models
- NLP & AI Orchestration:** LangChain, Langflow, NLP Pipelines, BeautifulSoup, Selenium, RestAPI, RAG pipelines, Vector Databases (FAISS, ChromaDB), Large Language Models (LLMs)
- Version Control:** Git, GitHub
- Data Visualization & BI:** Power BI, Tableau, Matplotlib, Seaborn, Streamlit
- Big Data & Engineering:** Hadoop, PySpark, ETL Pipelines
- Geospatial & Remote Sensing:** Google Earth Engine, QGIS

## CERTIFICATIONS

- Tata — GenAI Powered Data Analytics Job Simulation, Forage — Jun 2026
- Deloitte Australia — Data Analytics Job Simulation, Forage — Nov 2025
- EDA & Data Visualization in Data Science — Scaler, Mar 2025